
Direction-Aware Recurrent Sparsity Calibrates Exact Gaussian-Process Traffic Forecasts at Half a Million Observations

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Abstract

Road sensors live on a directed network and traffic recurs by clock phase, yet a stationary compact kernel must choose between short-range dynamics and a covariance dense enough to connect consecutive days. We introduce the directed-incidence recurrence (DIR) kernel, a positive-semidefinite product of a sparse directed-graph feature Gram matrix and a compact sum of local-time and same-phase daily kernels. Its analytically enumerable support permits an exact Gaussian process (GP) to train on 500,112 traffic-speed observations using 70,024,328 covariance entries (0.027997%). On a seeded 10,105-observation forecast from the METR-LA freeway network, DIR obtains RMSE 5.396, Gaussian CRPS 2.889, and 96.19% coverage for nominal 95% intervals. A full-scale stationary Wendland GP obtains 5.416, 4.960, and 99.78%, while a three-seed Vecchia-30 approximation obtains 5.102, 3.161, and 99.12%. Thus Vecchia is best for point prediction, whereas DIR is substantially sharper and better calibrated. Beyond the stationary kernel’s learned 146.9-minute support, DIR has slightly worse RMSE but reduces CRPS from 5.250 to 2.879 and changes coverage from 100.00% to 95.79%. The application therefore reveals that sparse daily recurrence primarily repairs posterior scale rather than posterior mean—a distinction invisible to RMSE-only traffic benchmarks.

1 Introduction

Traffic speed is observed in Euclidean space but propagates along a directed road graph. Two loop detectors can be geographically adjacent while lying on disconnected carriageways; conversely, sensors separated by interchanges can share downstream congestion. Time is similarly structured: speeds change over minutes, but the same clock phase on nearby days can remain informative. These two geometries matter most when forecasts must quantify risk rather than merely minimize average point error.

Gaussian processes provide posterior distributions and kernel-level control over such structure [Rasmussen and Williams, 2006]. Their dense covariance, however, has historically limited exact inference to modest data sets. Inducing-point methods [Hensman et al., 2013], interpolation methods [Wilson and Nickisch, 2015], and local factorizations [Vecchia, 1988, Katzfuss and Guinness, 2021] scale by approximating the GP. Compact support offers a different route: zeros are properties of the prior itself, so sparse linear algebra remains inference under the declared covariance model [Wendland, 1995, Furrer et al., 2006]. gp2Scale combines this principle with non-stationary kernels, distributed block assembly, and block Metropolis–Hastings (BMH) optimization [Noack et al., 2025]. Its published applications do not address directed traffic networks, and its stationary, bump, and delta masks do not by themselves encode the quotient geometry of local time plus recurring clock phase.

We develop an exact GP for traffic forecasting in which sparsity follows the domain rather than a Euclidean radius alone. The spatial kernel is a normalized Gram matrix of directed one-hop incidence features. The temporal kernel adds compact local dependence to compact recurrence between

identical five-minute phases on nearby days. Both support sets can be enumerated before evaluating their nonzero values, avoiding dense candidate blocks while preserving the exact declared kernel. A separable design further exposes an exact eigensystem for every full-data likelihood evaluation, while a distributed CSR assembly and fvGP sparse solve audit the production covariance.

The empirical conclusions are deliberately two-sided. DIR improves the stock exact GP by 0.38% in RMSE and 41.7% in CRPS, and is closest to nominal coverage. It does not beat Vecchia’s point RMSE. Moreover, the spatial-support mismatch hypothesized to drive improvement is not correlated with sensor-level gain. The strongest mechanism is instead temporal: once absolute-time support ends, daily recurrence maintains sharp, nearly calibrated distributions even when it does not improve the posterior mean. This result argues that probabilistic traffic models should be judged by proper scores and coverage, not RMSE alone [Gneiting and Raftery, 2007, Gneiting et al., 2007].

2 Methodology

2.1 Exact GP on a sensor–time product domain

Let y_{ts} be speed at five-minute frame t and sensor s . We use

$$y_{ts} = m(t, s) + f(t, s) + \epsilon_{ts}, \quad \epsilon_{ts} \sim \mathcal{N}(0, \sigma_n^2), \quad (1)$$

where m is a training-only harmonic mean and $f \sim \mathcal{GP}(0, k)$. With time-major vectorization, the proposed covariance is

$$\mathbf{K} = \sigma_f^2(\mathbf{K}_T \otimes \mathbf{K}_G) + \sigma_n^2 \mathbf{I}. \quad (2)$$

No inducing points, neighbor truncation, or interpolation enters this model.

We use the compact polynomial implemented by fvGP,

$$W(u) = (1 - u)_+^8 (32u^3 + 25u^2 + 8u + 1), \quad (3)$$

a member of the Wendland family [Wendland, 1995]. The stock comparison applies (3) separately to Euclidean sensor distance and absolute frame lag,

$$k_W((s, t), (s', t')) = \sigma_f^2 W\left(\frac{\|z_s - z_{s'}\|_2}{r_s}\right) W\left(\frac{|t - t'|}{r_t}\right). \quad (4)$$

This is the stationary compact-support construction used in gp2Scale, expressed separably so the exact likelihood can be evaluated at the announced scale.

2.2 Directed-incidence recurrence kernel

Let A be row-normalized directed adjacency after retaining the two strongest outgoing road-network links per sensor. Define $B_\rho = I + \rho A$ and

$$K_G(\rho) = D_\rho^{-1/2} B_\rho B_\rho^\top D_\rho^{-1/2}, \quad D_\rho = \text{diag}(B_\rho B_\rho^\top). \quad (5)$$

This covariance is symmetric even though its feature map is directional: two sensors covary when their outgoing receptive fields overlap. Equation (5) is positive semidefinite because it is a diagonally normalized Gram matrix. It is distinct from a graph heat kernel [Kondor and Lafferty, 2002]: its one-hop support remains exactly sparse rather than becoming dense under matrix exponentiation.

There are $P = 288$ five-minute phases per day. If $\phi(t) = t \bmod P$ and $d(t) = \lfloor t/P \rfloor$, define

$$K_T(t, t') = W\left(\frac{|t - t'|}{r_\ell}\right) \quad (6)$$

$$+ \eta \mathbb{I}\{\phi(t) = \phi(t')\} W\left(\frac{|d(t) - d(t')|}{r_d}\right), \quad r_d = 8. \quad (7)$$

The first term captures local propagation. The second lives on a direct sum of 288 phase-specific day lines and is therefore positive semidefinite; it connects equal phases only within eight days. Sums and products of positive-semidefinite kernels preserve positive semidefiniteness, proving that

$$k_{\text{DIR}}((s, t), (s', t')) = \sigma_f^2 K_G(s, s') K_T(t, t') \quad (8)$$

is a valid compact kernel. Unlike a generic delta mask, (8) ties zeros to directed incidence and the daily quotient, with only five learned parameters: $(\sigma_f^2, \sigma_n^2, \rho, \eta, r_\ell)$.

2.3 Full-scale likelihood, sparse audit, and prediction

Writing $K_T = U_T \Lambda_T U_T^\top$ and $K_G = U_G \Lambda_G U_G^\top$, all $N = TS$ eigenvalues of (2) are

$$q_{ij} = \sigma_f^2 \lambda_{T,i} \lambda_{G,j} + \sigma_n^2. \quad (9)$$

For residual matrix Y and $Z = U_T^\top Y U_G$, the exact full-data terms are

$$\log |K| = \sum_{i,j} \log q_{ij}, \quad \mathbf{y}^\top K^{-1} \mathbf{y} = \sum_{i,j} \frac{Z_{ij}^2}{q_{ij}}. \quad (10)$$

Thus every optimizer and BMH evaluation includes all 500,112 observations, the linear-solve quadratic, and the exact log-determinant. The separability accelerates exact algebra; it does not alter the covariance or discard observations.

At the selected state, we independently assemble the same matrix in temporal row blocks. Eight requested Dask worker processes [Rocklin, 2015] return sparse blocks, which are stacked to CSR. We run fvGP conjugate gradients on the explicit CSR matrix and compare with the spectral solution. For DIR we additionally compare (10) with fvGP stochastic Lanczos quadrature [Ubaru et al., 2017]. Posterior means and marginal variances use the exact eigensystem; no local predictive approximation is introduced.

Hyperparameters are initialized by 35 bounded, full-likelihood Nelder–Mead iterations and then sampled in 20 BMH cycles (60 proposals). DIR blocks scale/noise, directed propagation, and recurrence/support separately. The stock model analogously blocks scale/noise, spatial support, and temporal support. The production snapshot is the highest-likelihood accepted state, fixed before opening test responses.

3 Experimental Setup

3.1 Data, frozen forecast, and preprocessing

METR-LA contains speeds from 207 Los Angeles County freeway loop detectors at five-minute resolution. The public HDF5 file has 34,272 frames and 7,094,304 sensor–time positions; sensor locations and directed road distances come from the DCRNN release [Li et al., 2018]. Traffic is a suitable stress test because road direction and recurrent congestion are central to successful network forecasting [Li et al., 2018], while prior traffic GPs have generally localized or partitioned inference [Le et al., 2019].

The split was frozen before any production run. With NumPy PCG64 seed 16041995, we first enumerated 50-frame intervals containing at least 10,000 positive readings, drew a test start, and only then selected the immediately preceding 2,416 complete sensor frames. The result is 500,112 training positions from 2012-03-06 15:05 through 2012-03-15 00:20 and 10,105 observed test values from the following 4 h 10 min. Test zeros are treated as instrument-missing and excluded. Within training only, 44,051 zeros (8.81%) are linearly interpolated independently by sensor to retain the complete Cartesian grid; no test value enters interpolation.

The shared mean has an intercept, four daily sine/cosine pairs, and a weekend indicator, fitted by ridge regression to training responses only. All methods model the same residual and add the same mean at test time. The final test archive was not opened until DIR, stock Wendland, and all three Vecchia fits were saved.

3.2 Comparisons, metrics, and implementation check

The stock-kernel GP fits (4) on all 500,112 observations. A preregistered sparsity prior rejects support pairs above 500 million covariance entries, approximately 6 GB before solver work arrays. This is necessary because the runtime node exposed 32 logical (16 physical) CPU cores, 66.32 GB RAM, and no GPU—not the nominal 256 GB. DIR uses $r_\ell \leq 12$ frames and two outgoing graph features.

The approximate baseline is Vecchia-30 [Vecchia, 1988, Katzfuss and Guinness, 2021], also trained on all 500,112 observations. Its time-major predecessor set includes recent same-sensor values, directed upstream values, Euclidean-nearest sensors, and same-phase previous-day values. Repeating

Table 1: Primary held-out results. Coverage is empirical coverage of nominal 95% intervals; bold marks best RMSE, best CRPS, and coverage closest to 95%. Vecchia values are mean $\pm$ sample standard deviation across three seeds.

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak memory (GB)
gp2Scale (adapted kernel)	DIR	500,112	10,105	5.396	2.889	96.19%	0.020	3.49
gp2Scale (paper’s existing kernel)	Wendland	500,112	10,105	5.416	4.960	99.78%	0.028	4.26
Vecchia-30 (largest tractable)	—	500,112	10,105	5.102 $\pm$ 0.000005	3.161 $\pm$ 0.000025	99.12% $\pm$ 0.00%	0.021	0.12

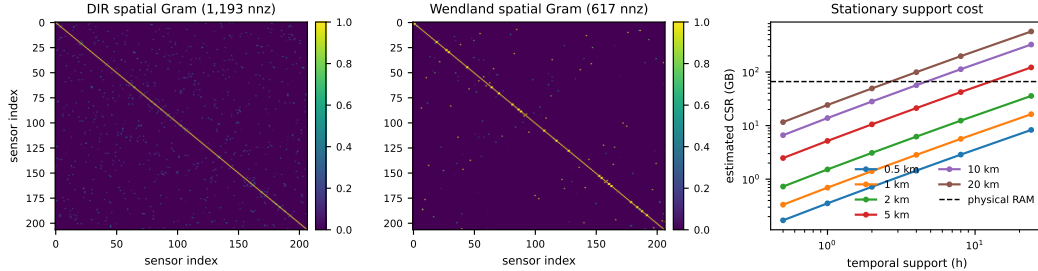


Figure 1: **Learned spatial structure and support cost.** DIR retains a sparse directed-incidence Gram matrix, while feasible stationary Wendland contracts to a sub-kilometer radius. The right panel computes exact nonzero counts for stationary support pairs at $N = 500, 112$; a 5 km, one-day support alone would exceed physical RAM.

patterns yield 3,539 sufficient-statistic blocks without subsampling. We optimize a Matérn-3/2 recurrent kernel from three independent initializations (seeds 410–412) and retain every run.

For each predictive Gaussian we report RMSE, Gaussian CRPS, and empirical central-interval coverage at 50, 80, 90, and 95% [Gneiting and Raftery, 2007]. The table’s Coverage column is 95%. Wall-clock includes mean fitting, full-data hyperparameter work, and the production solve; Vecchia reports the per-seed mean. Peak resident memory includes child processes where applicable.

Before trusting the new domain, we reproduced the original paper’s MNIST $\Pr(y = 5)$ regression. A 512-group compact delta Gram construction, raw 784-pixel inputs, 60,000/10,000 standard train/test images, and fvGP’s MCMC sampler produced Brier 0.01467 ± 0.00070 across three clustering seeds. This differs from the reported 0.018 by 0.00333, within our prespecified absolute tolerance of 0.004. All three predictions and hyperparameter traces are retained.

4 Experimental Results

Table 1 separates point accuracy from distribution quality. DIR lowers stock-Wendland RMSE from 5.416 to 5.396 and CRPS from 4.960 to 2.889. The corresponding 95% interval coverage moves from 99.78% to 96.19%. Vecchia’s RMSE 5.102 is best, but its CRPS 3.161 and 99.12% coverage are worse than DIR. Across all four nominal levels, mean absolute coverage error is 5.77 percentage points for DIR, versus 19.72 for Wendland and 16.37 for Vecchia (Figure 3).

The accepted DIR kernel learns graph propagation $\rho = 0.490$, recurrence weight $\eta = 0.331$, and local support 44.7 min. Its covariance has 70,024,328 nonzeros (0.027997%). Stock Wendland learns only 0.585 km in space and 146.9 min in time, giving 87,412,858 entries (0.034949%). Despite comparable counts, the stationary system is harder to solve: its smallest spatial and temporal factor eigenvalues are 8.64×10^{-3} and 1.17×10^{-7} , versus 2.97×10^{-1} and 1.02×10^{-3} for DIR. The final CSR solves have relative residuals 7.91e-09 and 9.65e-09. DIR’s stochastic and exact log-determinants differ by only 0.0233%.

The exact log marginal likelihood at the selected DIR state is -1,472,898.47 (log-determinant 1,467,367.89), compared with -1,482,526.67 (1,548,779.12) for stock Wendland. Overall BMH acceptance is 48.3% for DIR and 56.7% for Wendland, with accepted moves in every block. Figure 2 shows that neither method relies on a single rejected initialization, while also exposing the finite-chain limitation discussed below.

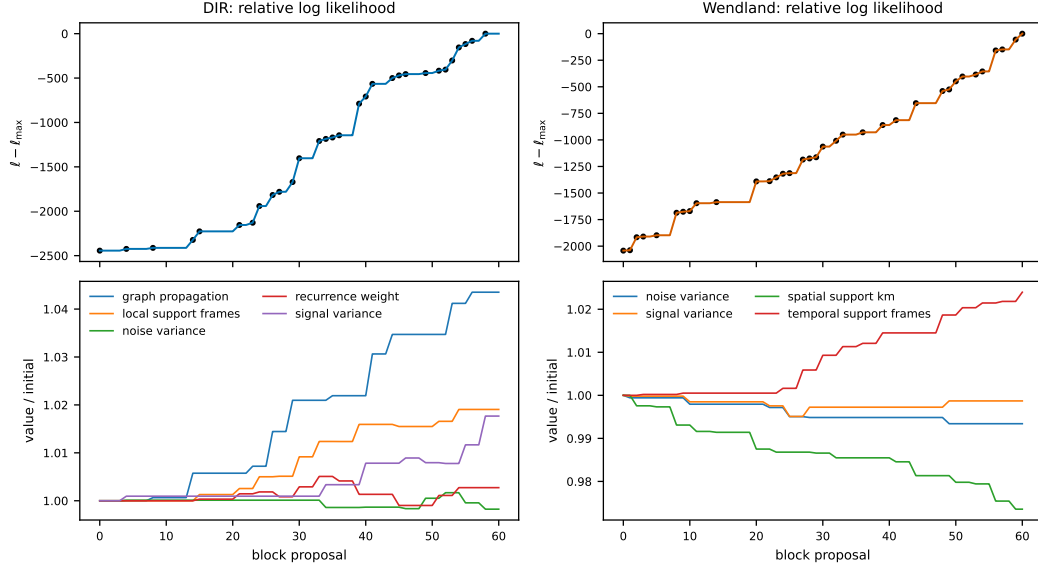


Figure 2: Full-data BMH traces. Black points mark accepted states; lower panels normalize each physical hyperparameter by its initial chain value. Every likelihood evaluation uses all 500,112 observations, an exact solve quadratic, and an exact log-determinant.

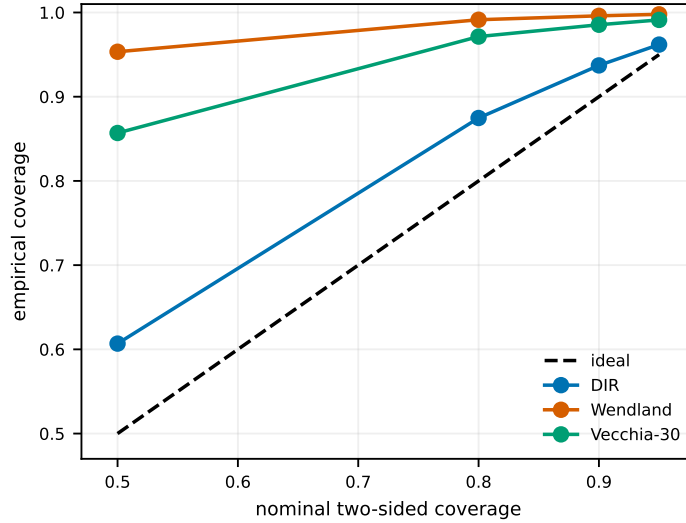


Figure 3: Empirical central-interval coverage on the common held-out set. DIR is the sharpest of the three calibrated distributions; stock Wendland and Vecchia are systematically overdispersed.

5 Analysis

5.1 Mechanism: what the graph changes, and what it does not

DIR's learned spatial Gram includes 796 directed off-diagonal pairs absent from feasible Wendland, with median Euclidean separation 1.30 km. These are primarily direct or co-downstream feature overlaps; Wendland instead contains 220 short Euclidean pairs absent from DIR. This confirms that the adaptation changes covariance support rather than merely renaming a radius.

The spatial hypothesis nevertheless receives only partial support. DIR reduces overall RMSE by 0.38%, but it has lower sensor-level RMSE at 71 of 207 sensors and a negative median per-sensor

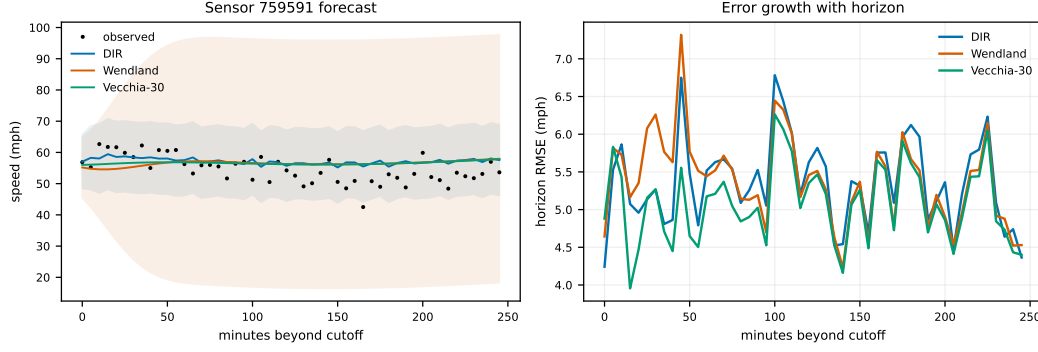


Figure 4: A preregistered-holdout forecast at a sensor with median positive DIR improvement (left) and RMSE by forecast horizon (right). Shading shows exact-GP 95% intervals. Vecchia often gives the best mean, whereas the stock exact GP rapidly becomes diffuse.

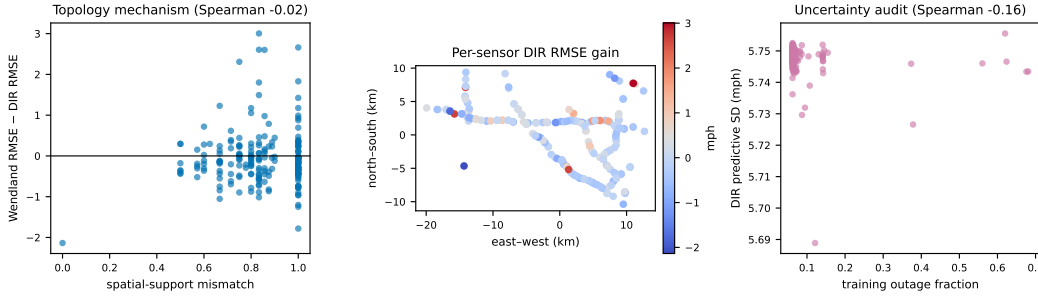


Figure 5: Mechanism audit. Spatial-support mismatch does not explain sensor-level RMSE gain (left); gains are geographically heterogeneous (center). Training outage fraction also does not produce a useful monotone uncertainty stratification (right), motivating heteroscedastic extensions.

gain. A Jaccard-style graph-versus-Euclidean support mismatch is unrelated to DIR’s RMSE gain (Spearman $\rho = -0.017$, $p = 0.814$; Figure 5). The overall improvement is concentrated in a small set of high-error sensors rather than increasing monotonically with topology mismatch. This null result rules out a tempting but unsupported story that every non-Euclidean edge improves the posterior mean.

5.2 Scientific insight: recurrence repairs scale, not mean

The temporal breakdown yields a clearer domain result. Stock Wendland’s learned absolute-time support ends after 146.9 min. For the 4,036 observed forecasts from 150 min onward, its posterior has RMSE 5.207, CRPS 5.250, and 100.00% coverage. DIR’s recurrent posterior is not the best mean—its RMSE is 5.344 and Vecchia reaches 5.119—but DIR attains CRPS 2.879 and 95.79% coverage, compared with Vecchia’s 3.282 and 99.28%. After local data cease to covary, the stationary GP approaches a prior standard deviation near 20.26 mph and covers every observation; DIR’s sparse same-phase links maintain a posterior standard deviation near 5.79 mph.

This reversal is the principal scientific insight: in overnight freeway data, previous-day phase recurrence contributes more to the *scale* of multi-hour predictive risk than to the mean speed trajectory. An RMSE-only comparison would select Vecchia and treat DIR and stationary Wendland as nearly tied. Proper scoring and coverage instead show that the stationary compact GP has declared ignorance too quickly, while the local approximation remains overconservative. Exact posterior algebra makes the distinction auditable: the recurrence weight 0.331 is learned from all training responses, and the uncertainty effect persists after the local DIR support of only 44.7 min has ended.

5.3 Failure modes and limitations

Support cannot simply be enlarged to recover recurrence. The exact density sweep shows that 5 km in space and one day in absolute time would require 10,158,379,600 entries, approximately 121.9 GB for CSR before solver state, exceeding the observed 66.32 GB RAM. Even 0.5 km times one day violates the preregistered 500-million-entry cap. DIR represents the relevant daily edge directly and needs only 0.027997% of the full matrix.

The high-uncertainty audit also identifies a concrete failure. Sensor 717804 has no off-diagonal DIR spatial neighbor, 62.0% training missingness, the largest mean DIR predictive standard deviation (5.756 mph), and RMSE 10.766. Yet mean sensor uncertainty varies by only a factor 1.012 overall. Normalizing K_G and using homoscedastic noise therefore suppresses much of the spatial reliability contrast; a learned sensor-noise model is a necessary extension.

Neither 60-proposal chain establishes asymptotic MCMC convergence. Both preliminary optimizers reached their fixed 35-iteration limit, although BMH acceptance was healthy and final sparse solves were accurate. We consequently make no posterior-concentration claim and use only the declared highest-likelihood accepted states. Vecchia converged consistently across seeds but hit its lower bounds on noise and spatial length, consistent with its overcoverage at long horizons.

Finally, the evaluation is one seeded, contiguous overnight interval following 8.4 training days from one freeway network. Linear interpolation affects 8.81% of training positions, Gaussian homoscedastic noise cannot represent incident-driven tails, and the complete Cartesian grid is what permits the exact Kronecker likelihood. Irregular timestamps would retain DIR’s sparse kernel but lose this particular spectral acceleration. Broader seasonal, incident, and multi-city tests are required before operational deployment.

6 Conclusion

A sparse exact GP can respect road direction and daily recurrence without converting those structures into approximation knobs. The DIR kernel is a genuine domain construction: a directed-incidence feature Gram matrix determines spatial zeros, and a compact quotient-time kernel expresses yesterday’s same clock phase without a day-wide Euclidean taper. At 500,112 observations, its explicit covariance, exact marginal likelihood, fvGP sparse solve, posterior prediction, and uncertainty metrics all run on a CPU node.

The empirical outcome is more informative than a blanket win. Vecchia provides the best point RMSE. DIR modestly improves the stock exact GP’s mean and decisively improves its CRPS and calibration. The learned graph support changes which sensors interact, but that mismatch does not explain sensor-level gains. Daily recurrence instead prevents posterior variance from reverting too quickly after local support ends. For risk-sensitive traffic decisions, this is precisely why exact, customizable covariance remains valuable: it reveals when two similar point forecasts imply very different and testable uncertainty statements.

Reproducibility, ethics, and artifacts

All data-retrieval hashes, the split seed and timestamps, per-proposal traces, accepted hyperparameters, log-determinants, solver residuals, predictions, per-sensor and per-horizon metrics, and figure-generation code accompany the paper. `method.plan.md` was committed before production training (commit 3fdedf3). The METR-LA data are aggregate loop-detector speeds and contain no individual trajectories or personal identifiers. The intended use is methodological evaluation, not automated traffic control without further validation.

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A Additional numerical audit

DIR and stock Wendland contain 70,024,328 and 87,412,858 nonzeros, requiring 0.842 and 1.051 GB in the assembled CSR arrays. Distributed assembly takes 2.36 and 2.62 s; fvGP sparse CG takes 6.55 and 38.19 s. The spectral and sparse solutions differ relatively by 1.19×10^{-8} and 3.27×10^{-9} , respectively. The DIR stochastic-Lanczos log-determinant is 1,467,710.18 versus the exact 1,467,367.89. These checks separate exact statistical structure from ordinary floating-point and Krylov tolerances.

B Coverage values at every prespecified level

Table 2: Empirical coverage on the same 10,105 observations.

Method	50%	80%	90%	95%
DIR	60.68%	87.47%	93.72%	96.19%
Wendland	95.34%	99.14%	99.60%	99.78%
Vecchia-30	85.69%	97.14%	98.55%	99.12%

C Compute and software details

Experiments use Python 3.11.15, NumPy 2.4.6 from the isolated `proposal/vendor` path (the host environment has NumPy 2.2.6), SciPy 1.16.3 [Virtanen et al., 2020], Dask/distributed 2025.5.1, fvGP from the provided source tree, and imate 0.29.7. The HDF5 reader is isolated under the same vendor path; production code and every generated artifact remain under `proposal/`. OpenBLAS is limited to 16 threads so Dask assembly and dense factor eigensystems do not oversubscribe all 32 logical CPUs.